Topic 5: FIR Filters

Summary based on J. H. McClellan, R. W. Schafer, and M. A. Yoder, Signal Processing First, Prentice Hall, 2003.

Topics: discrete-time systems, moving-average filters, general FIR filters, causality, unit impulses and impulse responses, convolution, block-diagram implementations, linearity, time invariance, and cascaded LTI systems.

> **Key idea.** An FIR filter forms each output sample as a finite weighted sum of present and delayed input samples. Its finite impulse response completely characterizes the system, and convolution provides the general rule for computing the response to any input.

5.1 Discrete-time systems

A discrete-time system is a computational process that transforms an input sequence into another sequence called the output signal.

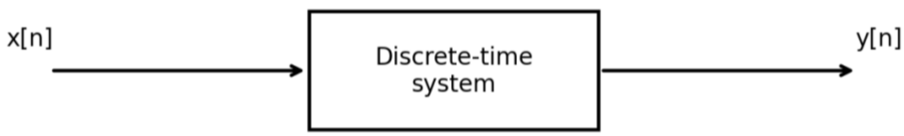

Figure 5.1. Block diagram of a discrete-time system.

In general, the operation of a system is denoted by

$$y[n] = \Gamma\{x[n]\}$$

A simple example is a system that scales the input by a constant:

$$y[n] = 2x[n]$$

5.2 Moving-average filter

A simple but useful transformation of a discrete-time signal is the moving average. Two or more consecutive samples are averaged to produce a new, smoother sequence. An FIR filter generalizes this idea. Averaging is useful when data fluctuate and should be smoothed before interpretation, for example in financial or measurement data.

Practical example. A long moving-average filter can reveal a slow trend hidden by rapid fluctuations. The textbook applies a 51-point causal averager to weekly stock-market data. Each output sample is the average of the current value and the previous 50 values:

$$y[n] \;=\; (1/51)\, \Sigma\text{\_}(k = 0)\text{\^{}}50\; x[n-k]$$

The result is much smoother, but the causal filter also introduces a delay of approximately 25 samples. A centred moving average removes this visible delay when future samples are available.

Consider the three-point moving average of the finite triangular sequence shown in Figure 5.2.

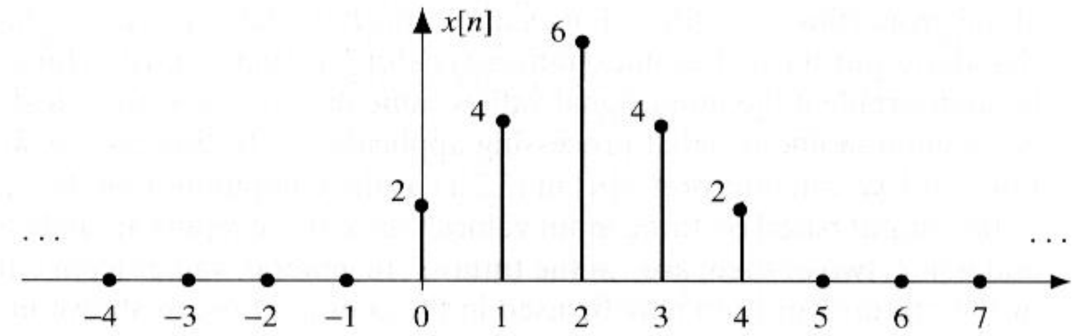

Figure 5.2. Finite-length input signal, x[n].

The output is calculated as

$$y[n] = \frac{1}{3}(x[n+1] + x[n] + x[n-1])$$

More generally, an L-point moving average can be written as

$$y[n] = \frac{1}{L}\sum_{k=0}^{L-1} x\,[n-k]$$

This expression is called a difference equation. It completely describes the FIR system because it allows the output to be calculated for every index n.

n	n<−2	−2	−1	0	1	2	3	4	5	n>5
x[n]	0	0	0	2	4	6	4	2	0	0
y[n]	0	2/3	2	4	14/3	4	2	2/3	0	0

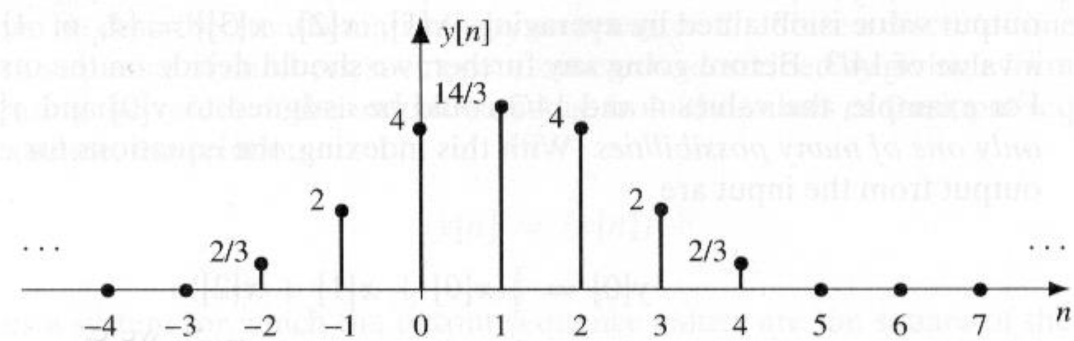

Figure 5.3. Output of the moving-average filter, y[n].

A filter that uses only present and past input values is causal: the output does not depend on data that have not yet arrived. A filter that uses future values is noncausal and cannot be implemented in real-time applications without delay.

A causal three-point moving average is

$$y[n] = \frac{1}{3}(x[n] + x[n-1] + x[n-2])$$

Worked example. For the finite sequence in Figure 5.2, evaluate the causal three-point average at n = 4. The filter uses the present sample and the two preceding samples:

$$y[4] = \frac{x[4] + x[3] + x[2]}{3} = \frac{2+4+6}{3} = 4$$

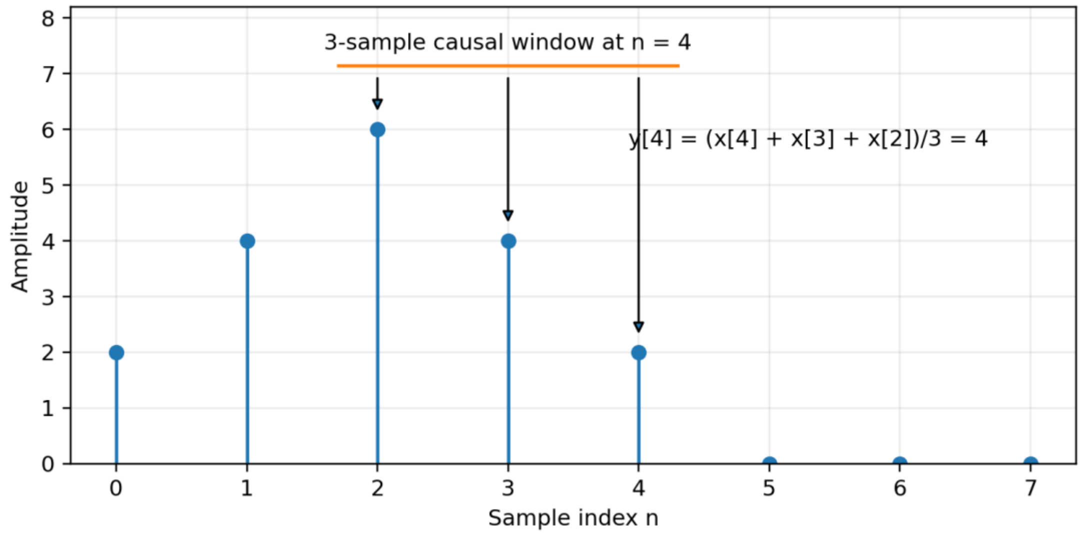

Figure 5.4. Sliding-window calculation for a causal three-point moving-average filter.

5.3 General FIR filter

The previous causal filter can be generalized to

$$y[n] = \sum_{k=0}^{M} b_k \, x[n-k]$$

The parameter M is the order of the FIR filter. The number of coefficients is the filter length L, so

Worked example: coefficients and order. Let the FIR coefficients be b[0]=3, b[1]=−1, b[2]=2, and b[3]=1. The filter has order M=3 and length L=4, and its difference equation is

$$y[n] \;=\; 3x[n] \;-\; x[n-1] \;+\; 2x[n-2] \;+\; x[n-3]$$

This example makes clear that the coefficient list, the difference equation, and the impulse response are three equivalent descriptions of the same FIR filter.

$$L = M + 1$$

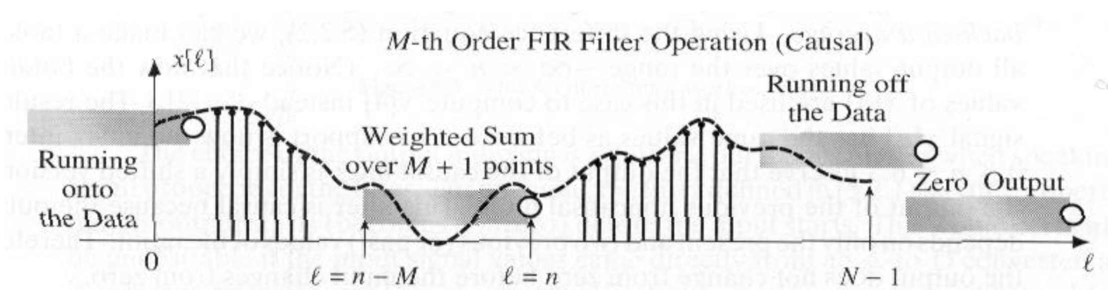

Figure 5.5. Operation of an order-M causal FIR filter as the processing window moves across the input.

5.3.1 Illustration of FIR filtering

Consider an input formed by a slowly increasing component plus a sinusoidal disturbance:

$$x[n] = (1.02)^n + \cos\left(\frac{2\pi n}{8} + \frac{\pi}{4}\right), \qquad 0 \le n \le 40$$

Treat the sinusoidal component as noise that should be reduced. For an order-2 causal moving-average filter,

$$y[n] = \frac{1}{3}\sum_{k=0}^{2} x\,[n-k]$$

The middle plot in Figure 5.6 shows that part of the sinusoidal fluctuation is removed, although the underlying trend is not recovered perfectly. Averaging over more samples gives stronger smoothing. With an order-6 filter,

$$y[n] = \frac{1}{7}\sum_{k=0}^{6} x\,[n-k]$$

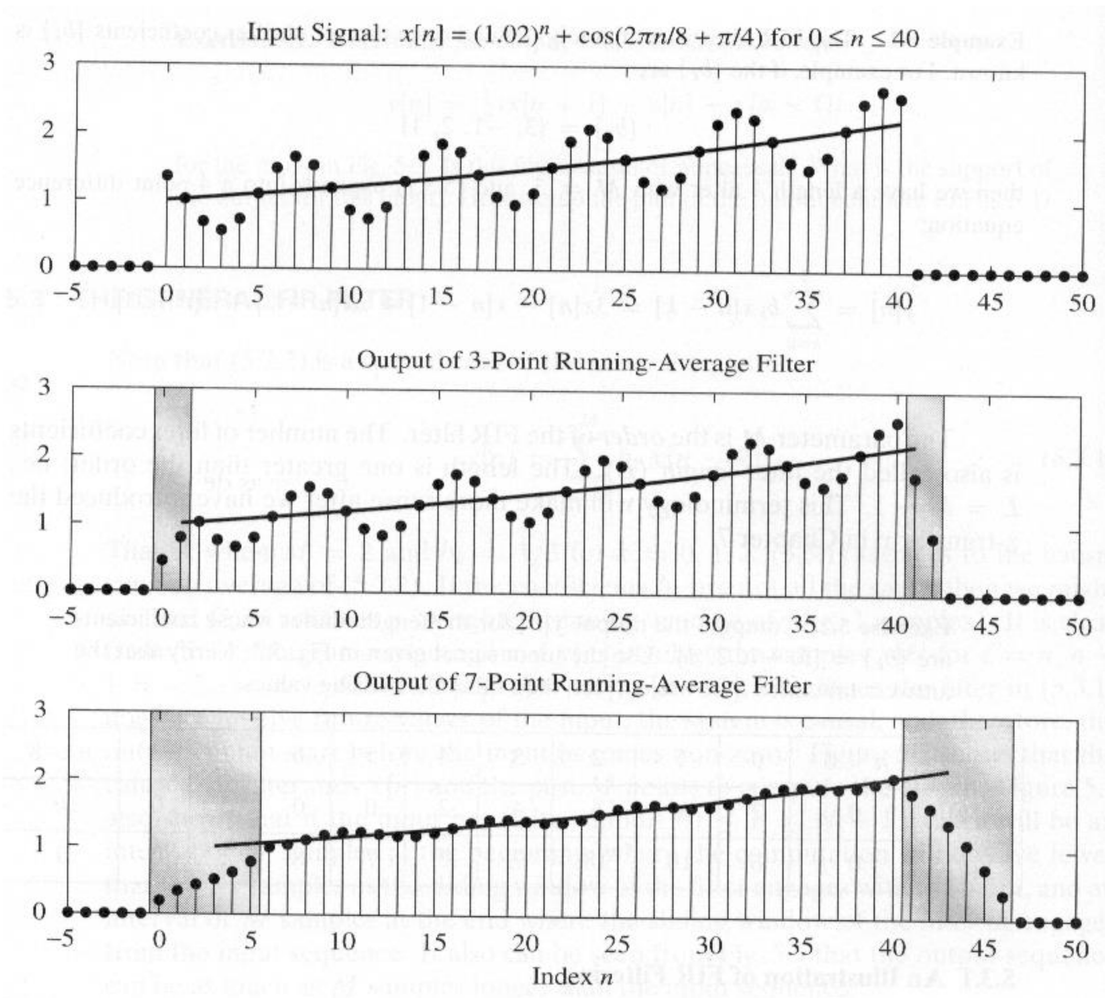

Figure 5.6. Moving-average filtering with 3-point and 7-point windows.

5.3.2 Unit impulse response

This section introduces three related ideas: the unit impulse sequence, the unit impulse response, and the convolution sum. The impulse response completely characterizes an FIR filter, and convolution provides a formula for calculating the output for any input when the impulse response is known.

5.3.2.1 Unit impulse sequence

The unit impulse is the simplest discrete-time sequence because it is nonzero only at n=0. It is denoted by the Kronecker delta:

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

A shifted impulse δ[n−3] is nonzero when n−3=0, that is, at n=3.

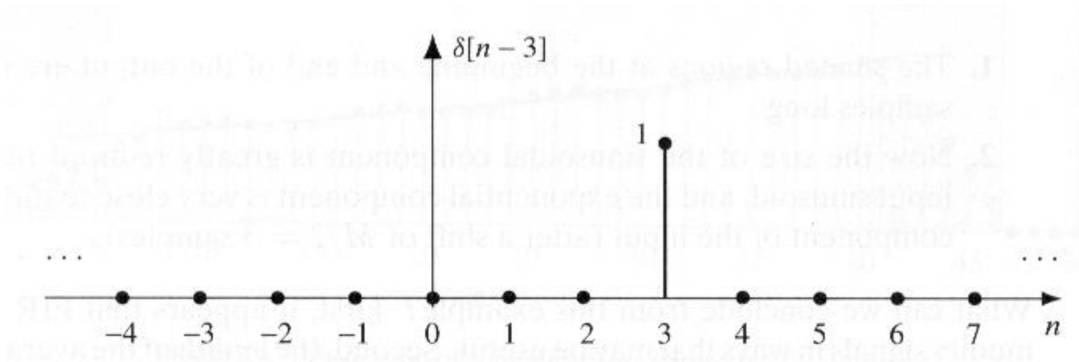

Figure 5.7. Shifted unit impulse sequence, δ[n−3].

Any sequence can be represented as a sum of scaled and shifted impulses. For the sequence in Figure 5.2,

$$x[n] = 2\delta[n] + 4\delta[n-1] + 6\delta[n-2] + 4\delta[n-3] + 2\delta[n-4]$$

In general,

$$x[n] = \sum_{k=-\infty}^{\infty} x\,[k]\delta[n-k]$$

5.3.2.2 Unit impulse response

When the input to an FIR filter is the unit impulse δ[n], the output is, by definition, the unit impulse response h[n], usually called simply the impulse response.

Figure 5.8. Block diagram defining the impulse response.

Substituting the unit impulse into the FIR difference equation gives

$$h[n] = \sum_{k=0}^{M} b_k\,\delta[n-k]$$

Only one term is nonzero for each value of n, so the impulse response is simply the sequence of filter coefficients:

$$h[n] = \begin{cases} b_n, & 0 \leq n \leq M \\ 0, & \text{otherwise} \end{cases}$$

Its duration is finite, which is why the system is called a finite impulse response filter.

Example: first-difference filter. The filter

$$y[n] \;=\; x[n] \;-\; x[n-1]$$

$$h[n] = \delta[n] - \delta[n-1]$$

responds to changes rather than constant levels. For a unit-step input u[n], all samples cancel except at the transition, so y[n]=δ[n]. This illustrates how a simple FIR filter can detect an onset or edge.

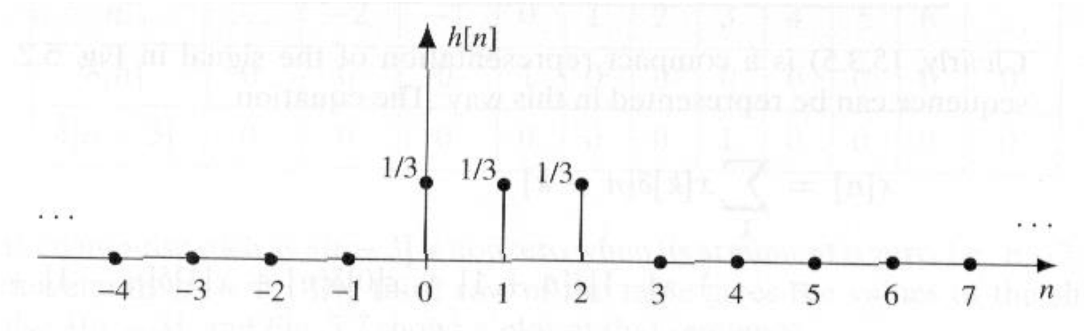

Figure 5.9. Impulse response of an order-2 moving-average filter.

5.3.2.3 Unit-delay system

An important elementary system delays, or shifts, the input by an integer number of samples:

$$y[n] = x[n - n_0]$$

For n_0=1 it is called a unit delay. A delay is the simplest FIR filter because all coefficients are zero except one. For a delay of three samples,

$$h[n] = \delta[n-3]$$

5.3.3 Convolution and FIR filters

Because the FIR coefficients are identical to the impulse-response samples, the general FIR equation can be written as

$$y[n] = \sum_{k=0}^{M} h[k]x[n-k]$$

This is the finite convolution sum. The output is said to be obtained by convolving x[n] and h[n], written

Worked convolution example. For x[n]={2,4,6,4,2} and h[n]={3,−1,2,1}, the convolution is obtained by scaling and shifting the input once for each impulse-response sample and summing the aligned rows:

$$y[n] = x[n] * h[n]$$

The resulting output sequence is y[n]={6,10,18,16,18,12,8,2}. The output length is 5+4−1=8 samples.

	0	1	2	3	4	5	6	7
h[0]x[n−0]	6	12	18	12	6	0	0	0
h[1]x[n−1]	0	-2	-4	-6	-4	-2	0	0
h[2]x[n−2]	0	0	4	8	12	8	4	0
h[3]x[n−3]	0	0	0	2	4	6	4	2
y[n]	6	10	18	16	18	12	8	2

Figure 5.10. Numerical convolution as a sum of scaled and shifted input sequences.

5.4 FIR-filter implementation

The FIR equation shows that implementation requires three operations: multiplying delayed input samples by the filter coefficients, adding the scaled values, and generating delays of the input sequence. Block diagrams provide a useful representation of these operations.

5.4.1 Basic blocks

The basic elements are a multiplier, an adder, and a unit-delay operator.

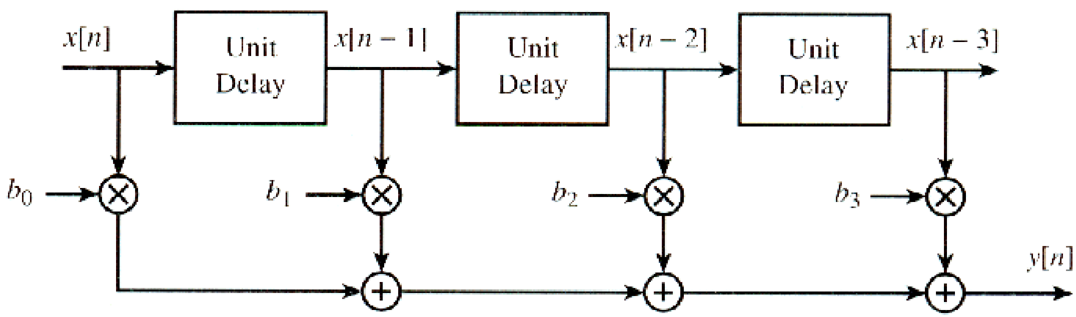

Figure 5.11. Basic blocks for constructing a discrete-time linear time-invariant system.

5.4.2 Block diagrams

Block-diagram notation specifies the interconnections among the basic elements and is especially useful for hardware structures. Figure 5.12 shows the direct-form realization of a third-order FIR filter.

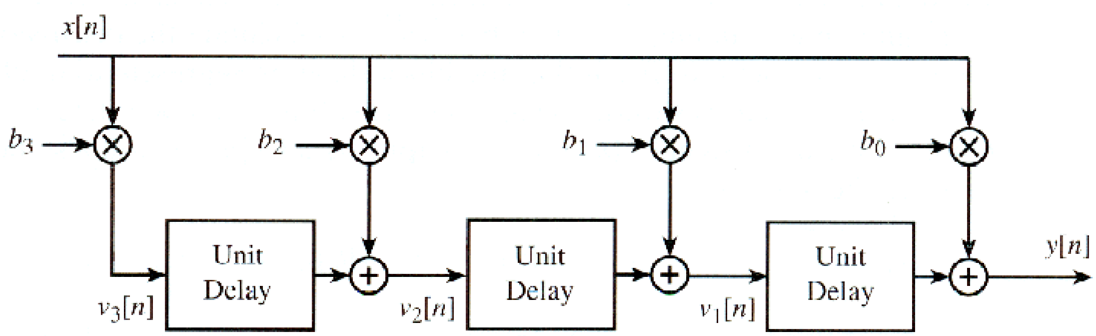

Figure 5.12. Direct-form block diagram of a third-order FIR filter.

The structure implements

$$y[n] = b_0x[n] + b_1x[n-1] + b_2x[n-2] + b_3x[n-3]$$

5.4.2.1 Alternative block diagrams

Many block diagrams can implement the same FIR filter, meaning that their external input-output behavior is identical. Figure 5.13 shows the transposed form of the same third-order filter.

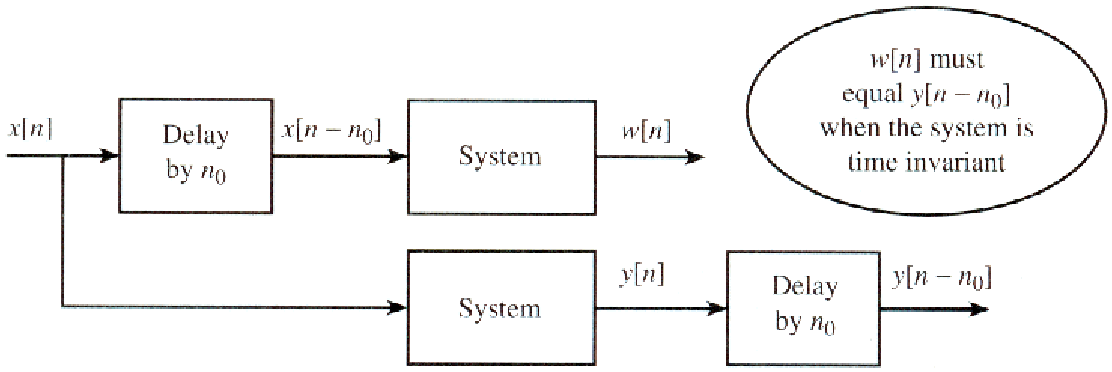

Figure 5.13. Transposed-form realization of the filter in Figure 5.12.

5.5 Linear time-invariant systems (LTI)

Linearity and time invariance simplify mathematical analysis and provide a clearer understanding of system behavior.

5.5.1 Time invariance

A discrete-time system is time invariant if shifting the input by n_0 samples shifts the output by exactly the same amount. If

$$y[n] = \Gamma\{x[n]\}$$

then time invariance requires

$$\Gamma\{x[n-n_0]\} = y[n-n_0]$$

for every integer n_0.

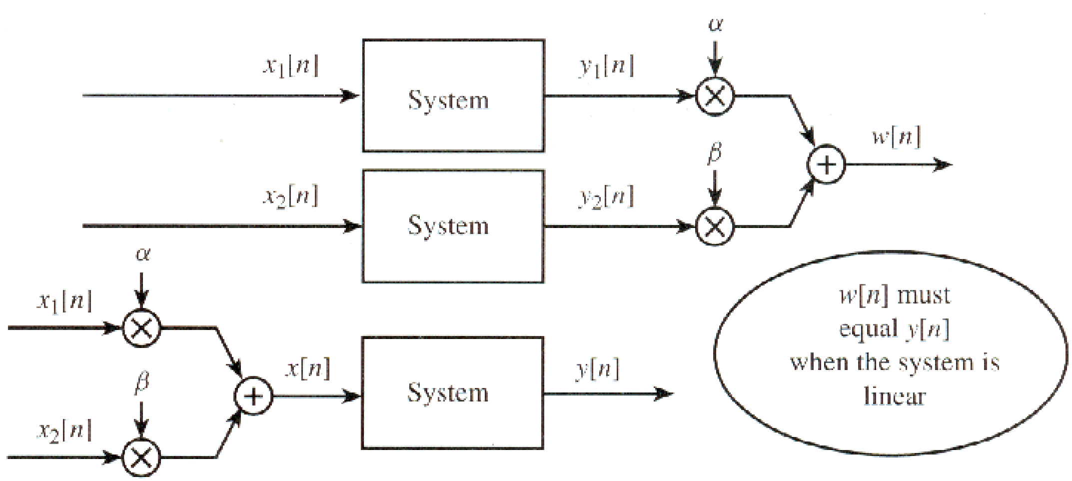

Figure 5.14. Test of the time-invariance property.

5.5.2 Linearity

A system is linear if, whenever

$$y_1[n] = \Gamma\{x_1[n]\}, \qquad y_2[n] = \Gamma\{x_2[n]\}$$

then, for any constants α and β,

$$\Gamma\{\alpha x_1[n] + \beta x_2[n]\} = \alpha y_1[n] + \beta y_2[n]$$

This is the superposition principle.

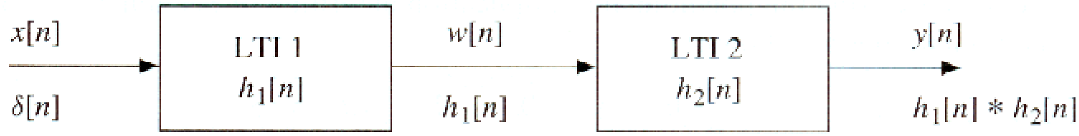

Figure 5.15. Test of the linearity property.

5.5.3 The FIR case

The general FIR system is both time invariant and linear. For time invariance, delaying the input gives

$$\Gamma\{x[n-n_0]\} = \sum_{k=0}^{M} b_k\, x[n-n_0-k]$$

whereas delaying the original output gives

$$y[n-n_0] = \sum_{k=0}^{M} b_k\, x[n-n_0-k]$$

The expressions are identical. Linearity follows directly because the summation distributes over scaled sums:

$$\Gamma\{\alpha x_1[n] + \beta x_2[n]\} = \alpha\Gamma\{x_1[n]\} + \beta\Gamma\{x_2[n]\}$$

5.6 Convolution and LTI systems

The impulse response is a complete characterization of an LTI system.

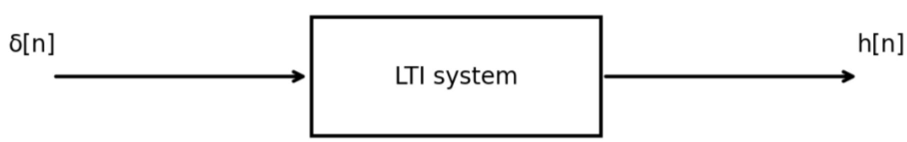

Figure 5.16. LTI system with an impulse input.

5.6.1 Derivation of the convolution sum

Write the input as a linear combination of scaled, shifted impulses:

$$x[n] = \sum_{k=-\infty}^{\infty} x\,[k]\delta[n-k]$$

By definition, the response to δ[n] is h[n]. Time invariance gives the response to a shifted impulse:

$$\Gamma\{\delta[n-k]\} = h[n-k]$$

Using linearity, the response to the complete input is

$$y[n] = \sum_{k=-\infty}^{\infty} x\,[k]h[n-k]$$

Therefore every LTI system can be represented by the convolution sum

$$y[n] = x[n] * h[n]$$

5.6.2 Properties of LTI systems

The algebraic properties of convolution correspond directly to properties of LTI systems.

5.6.2.1 Convolution as an operator

Convolution of x[n] and h[n] is denoted by *:

$$y[n] = x[n] * h[n]$$

Convolution with an impulse leaves a sequence unchanged:

$$x[n] * \delta[n] = x[n]$$

Convolution is commutative and associative:

$$x[n] * h[n] = h[n] * x[n]$$
$$(x[n] * h_1[n]) * h_2[n] = x[n] * (h_1[n] * h_2[n])$$

5.7 Cascaded LTI systems

Two LTI systems connected in cascade can be implemented in either order because convolution is commutative. Their overall impulse response is

Worked cascade example. Suppose the first system has $h_1[n]=\{1,1,1,1\}$ and the second has $h_2[n]=\{1,1,1\}$. The equivalent impulse response is the convolution

$$h[n] \ = \ h_1[n] \ * \ h_2[n]$$
$$h[n] \ = \ \{1,2,3,3,2,1\}$$

The same result is obtained whichever system is placed first. A single FIR filter with these six coefficients therefore reproduces the complete cascade.

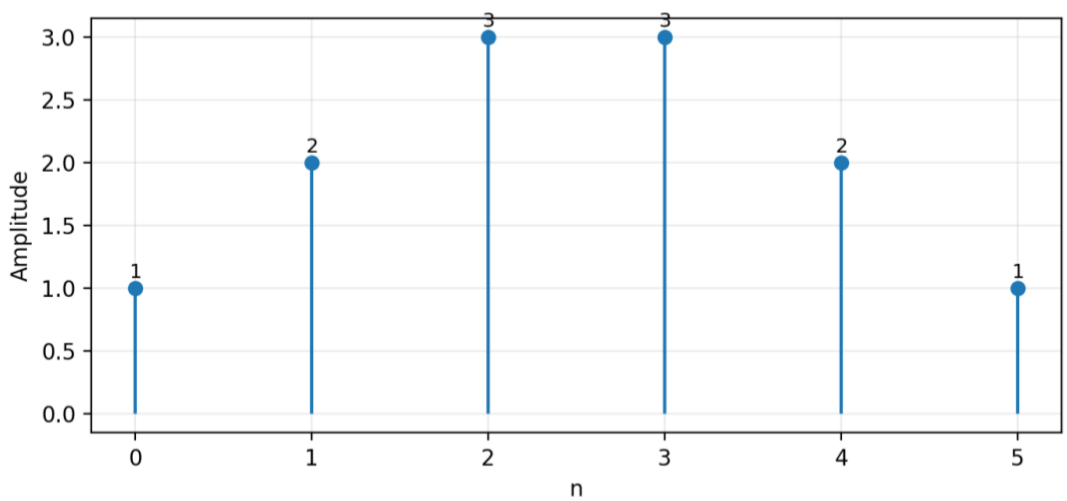

Figure 5.17. Equivalent impulse response of two cascaded moving-sum FIR filters.

$$h[n] = h_1[n] * h_2[n]$$

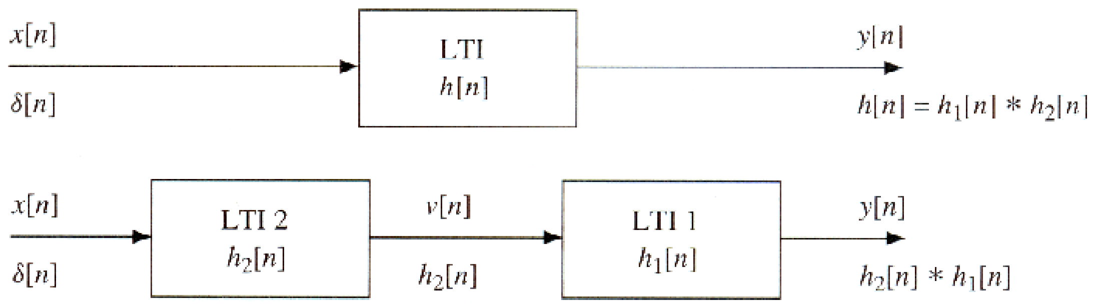

Figure 5.18. Two cascaded LTI systems and the equivalent system obtained by interchanging their order.

5.8 Notation

Symbol	Meaning
x[n]	Input discrete-time sequence
y[n]	Output discrete-time sequence
b_k	FIR filter coefficient at delay k
M	Filter order; the largest delay in an order-M FIR filter
L	Filter length, L = M + 1
h[n]	Impulse response of the system
δ[n]	Unit impulse (Kronecker delta)
u[n]	Unit-step sequence
n_0	Integer time shift or delay
*	Convolution operator
T{·}	System transformation operator
α, β	Arbitrary scalar constants used in the linearity test
$h_1[n]$, $h_2[n]$	Impulse responses of cascaded LTI systems